LESSON 5.9

Area of Regular Polygons

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a regular polygon by decomposing it into triangles.

WHY THIS MATTERS Your architecture firm is designing a hexagonal skylight for the lobby of a new museum. The glass fabricator needs to know the total area so they can cut the correct amount of glass. Each side of the hexagon is 6 feet, and the height of each triangle formed from the center is 5.2 feet. The height is rounded to the nearest tenth of a foot, so the area is approximate.

REMEMBER

Area of Regular Polygons

Area of Regular Polygons. Formula: Area = (Number of Sides) × (Area of One Triangle) A regular polygon is a closed shape with straight sides where every side is the same length and every angle is the same size. We can decompose it (break it apart) into equal triangles from the center, then add up their areas.

1. Decompose the regular polygon from center into n congruent triangles.
2. Find the area of one triangle using A = ½ × side × apothem (height).
3. Multiply by the total number of sides

Words: Regular Polygon *(Polígono regular)* · Decompose *(Descomponer)* · Triangle *(Triángulo)* · Composite *(Compuesto)* · Formula *(Fórmula)*

1 **Circle the letter of the best answer. Show how you know.**

How many triangular faces does a square pyramid have?

(A) 3 (B) 4

(C) 5 (D) 6

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

To find the surface area of a pyramid you add…

(A) the area of the base and the areas of all the triangular faces

(B) the volume and the base area

(C) the base perimeter and the height

(D) only the triangular faces

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A triangular pyramid has a base area of 20 in² and three faces of 14 in² each. What is its surface area?

(A) 34 in²

(B) 54 in²

(C) 62 in²

(D) 62 in³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

A square pyramid has a base 6 cm by 6 cm and four triangular faces of 15 cm² each. What is its surface area?

(A) 51 cm²

(B) 60 cm²

(C) 96 cm²

(D) 96 cm³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

The parks department hired your firm to finish a regular octagonal gazebo at the center of Patterson Park. Seen from above, the gazebo floor decomposes into 8 congruent triangles that meet at the center point, and each triangle has a base of 5 feet and a height of 6 feet. The flooring contractor charges $3 per square foot and needs a total price before Friday's budget meeting. The central-triangle height is rounded to the nearest foot. Estimate the area and price with that height. What is the area of one center triangle?

(A) 7.5 sq ft
(B) 11 sq ft
(C) 15 sq ft
(D) 30 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

6 **Circle the letter of the best answer. Show how you know.**

The parks department hired your firm to finish a regular octagonal gazebo at the center of Patterson Park. Seen from above, the gazebo floor decomposes into 8 congruent triangles that meet at the center point, and each triangle has a base of 5 feet and a height of 6 feet. The flooring contractor charges $3 per square foot and needs a total price before Friday's budget meeting. The central-triangle height is rounded to the nearest foot. Estimate the area and price with that height. What is the total floor area of the octagon?

(A) 15 sq ft
(B) 40 sq ft
(C) 120 sq ft
(D) 360 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

EXPLAIN YOUR THINKING

The hexagon and the octagon both have triangles, but the totals are different. What two things decide the total area of a regular polygon?

SENTENCE STARTER The total area is ___ because one triangle is ___ and there are ___ triangles.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Each tile decomposes into congruent triangles from its center. Work out the area of ONE triangle, then sort the tile. Use the given heights; nonsquare heights are rounded, so their areas are estimates.

Groups: One triangle ≤ 12 sq ft | One triangle > 12 sq ft

_____ Square tile: base 4 ft, height 2 ft

_____ Hexagon tile: base 4 ft, height ≈ 3.5 ft

_____ Octagon tile: base 6 ft, height ≈ 7.2 ft

_____ Pentagon tile: base 8 ft, height ≈ 5.5 ft

2 **Write the letter of the matching item on each line.**

Match each polygon tile to its total area. Use the given heights; nonsquare heights are rounded, so their areas are estimates.

_____ Square tile: 4 triangles, each ½×9×4.5

_____ Octagon tile: 8 triangles, each about ½×3×3.6

_____ Hexagon tile: 6 triangles, each about ½×5×4.3

_____ Equilateral triangle tile: 1 whole triangle, about ½×6×5.2

(A) 81 sq ft

(B) About 43.2 sq ft

(C) About 64.5 sq ft

(D) About 15.6 sq ft

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

The parks department hired your firm to finish a regular octagonal gazebo at the center of Patterson Park. Seen from above, the gazebo floor decomposes into 8 congruent triangles that meet at the center point, and each triangle has a base of 5 feet and a height of 6 feet. The flooring contractor charges $3 per square foot and needs a total price before Friday's budget meeting. The central-triangle height is rounded to the nearest foot. Estimate the area and price with that height. What will the flooring cost?

(A) $45

(B) $120

(C) $360

(D) $960

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

4 **Circle the letter of the best answer. Show how you know.**

A regular hexagonal floor tile splits from its center into 6 congruent triangles, each with an area of 12 cm². What is the total area of the tile?

(A) 12 cm²

(B) 18 cm²

(C) 36 cm²

(D) 72 cm²

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

5 Circle the letter of the best answer. Show how you know.

A regular pentagonal tile splits into 5 congruent triangles. Each triangle has a base of 4 in and a height of about 2.8 in. What is the total area of the tile? Use the given heights; nonsquare heights are rounded, so their areas are estimates.

(A) 5.6 in²
(B) 11.2 in²
(C) 28 in²
(D) 56 in²

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

6 Circle the letter of the best answer. Show how you know.

An octagon tile splits into equal triangles, each with base 6 ft and height ≈ 7.2 ft. How many triangles, and what is the total area? Use the given heights; nonsquare heights are rounded, so their areas are estimates.

(A) 21.6 sq ft
(B) 129.6 sq ft
(C) 151.2 sq ft
(D) 172.8 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

EXPLAIN YOUR THINKING

The hexagon and the octagon both have triangles, but the totals are different. What two things decide the total area of a regular polygon?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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